

Aukosh S. Jagannath

Curriculum Vitae

Department of Statistics and Actuarial Science
University of Waterloo
Mathematics 3 (M3) Rm 2114
200 University Ave W
Waterloo, ON, Canada
E-mail: a.jagannath@uwaterloo.ca
Homepage: <http://aukosh.github.io>

Education

Ph. D. in Mathematics, New York University, May 2016
Title: Variational and structural methods in mean field spin glasses

B.A. in Mathematics and Physics, New York University, May 2011
Summa Cum Laude with highest honors in Mathematics and highest honors in Physics

Academic Positions

University of Waterloo

Canada Research Chair in Mathematical Foundations of Data Science May 2023 -
Associate Professor, Department of Statistics and Actuarial Science Jul 2024 -
Assistant Professor, Department of Statistics and Actuarial Science Jul 2019 - Jun 2024
Cross-appointments: Department of Applied Mathematics, Cheriton School of Computer Science
External affiliations: Perimeter Institute (Affiliate)
(Leaves: Parental leave, Summer-Fall 2020; Parental leave, Summer 2022; Parental leave, Summer 2024)

Harvard University

Benjamin Pierce Fellow, Department of Mathematics, Sep 2016-Jun 2019 (on leave AY 16-17)
NSF MSPRF Postdoctoral Fellow, Department of Mathematics Sep 2017-Jun 2019

University of Toronto

NSF MSPRF Postdoctoral Fellow, Department of Mathematics, Sep 2016-Aug 2017

Research Interests

Probability, Mathematical Foundations of Data Science, Mathematical Physics

Awards and Honors

Outstanding Performance Award, University of Waterloo, 2025
Canada Research Chair Tier II 2023-2028
Outstanding Paper Award, NeurIPS 2022
Golden Jubilee Research Excellence Award, University of Waterloo, 2022
NSF Mathematical Sciences Postdoctoral Research Fellowship, 2016-2019
NSF Graduate Research Fellowship, 2011-2016
Dean's Dissertation Fellowship, NYU GSAS, 2015-2016
Wilhelm Magnus memorial prize, NYU Courant, 2015
College of Arts and Sciences Dean's Award for Scholarship, NYU CAS, 2011
Perley Lenwood Thorne Medal in Mathematics, NYU Courant, 2011
George Granger Brown Scholarship for Physics, NYU CAS, 2010-2011
Phi Beta Kappa (US National Honours Society)

Grant record

Total: ~ \$2 million CAD in funding as PI or Fellowship (incl. declined awards)

Natural Sciences and Engineering Research Council of Canada, Discovery Grant RGPIN-2026-04888 (PI), \$300,000 CAD 2026-2031
Ontario Research Fund (PI), \$ 80,000 CAD, 2024-2026
Canada Foundation for Innovation, John Evans Leaders' Fund, (PI), \$80,000 CAD, 2024-2026
University of Waterloo, Mathematics Faculty Research Fund (PI), 2023-2028, \$250,000 CAD
Canada Research Chairs Program, Federal Research Fund (PI), \$100,000 CAD, 2023-2028
Canada Research Chairs Program, Canada Research Chair Tier II (PI), \$500,000 CAD , 2023-2028
Natural Sciences and Engineering Research Council of Canada, Discovery Grant RGPIN-2020-04597 (PI) \$198,000 CAD 2020-2026
Natural Sciences and Engineering Research Council of Canada, Discovery Launch Supplement, DGECR-2020-00199 (PI), \$12,500 CAD 2020-2022
National Science Foundation (US) DMS 1854406 (PI), \$150,000 USD, (*Declined* due to move to Waterloo)
National Science Foundation (US), Mathematical Sciences Postdoctoral Research Fellowship, OISE-1604232 (PI), \$150,000 USD, 2016-2019
National Science Foundation (US), Graduate Research Fellowship, \$99,000 USD 2011-2016

Research

Preprints and Publications

(Preprints Available at <http://math.uwaterloo.ca/~a3jagann/research.html>)

The following use the alphabetical authorship convention common to theoretical fields unless starred ().*

44. C. Grant, A. Jagannath, J. Ko, Pseudo-maximum likelihood theory for high-dimensional rank one inference, *Annals of Statistics* Vol. 54, No. 4 pp. 2031-2054 (2026)
43. G. Ben Arous, R. Gheissari, J. Huang, and A. Jagannath, Local geometry of high-dimensional mixture models: Effective spectral theory and dynamical transitions, *Annals of Statistics* Vol. 54, No. 4, 1682-1711 (2026)
42. R. Gheissari and A. Jagannath, Universality of high-dimensional scaling limits of stochastic gradient descent, *Conference on Learning Theory* (COLT 2026), arXiv:2512.13634
41. A. Jagannath, T. Jones-McCormick, V. Sarangian, High-dimensional limit theorems for SGD: Momentum and Adaptive Step-sizes, *International Conference on Learning Representations 14* (ICLR 2026) , arXiv:2511.03952
40. *K. Ramsay, A. Jagannath, and S. Chenouri, Differentially private multivariate medians, *Journal of Machine Learning Research*, Vol. 26, No. 255 pp. 1-52 (2025) (authorship: equal, reverse seniority)
39. D. Gamarnik, A. Jagannath, and E. Kizildag, Shattering in the Ising p -spin model. *Probability Theory and Related Fields* Vol. 193, pp. 89-141 (2025)
38. G. Ben Arous, R. Gheissari, J. Huang, and A. Jagannath, Spectral alignment of stochastic gradient descent for high-dimensional classification tasks, *Annals of Applied Probability*, Vol. 35, No. 4, pp. 2767-2822 (2025)
(Conference version "High-dimensional SGD aligns with emerging outlier eigenspaces" appeared at *International Conference on Learning Representation 12* (ICLR 2024) (spotlight, top 5% of submissions), (2024))
37. *T. Jones-McCormick, A. Jagannath, S. Sen, Provable Benefits of Unsupervised Pre-training and Transfer Learning via Single-Index Models, *International Conference on Machine Learning 42* (ICML 2025) (spotlight, top 2.6% of submissions) (2025)
36. R. Gheissari, A. Jagannath, Y. Xu, Finding planted clique using gradient descent, *SIAM Journal on Mathematics of Data Science*, Vol. 7, No. 2, pp. 643–669 (2025)
35. *K. Ramsay, A. Jagannath, and S. Chenouri, An elementary concentration bound for Gibbs measures arising in statistical learning theory, *Transactions of Machine Learning Research 2025* (authorship: equal, reverse seniority)
34. A. Jagannath and P. Lopatto, Existence of the free energy of heavy-tailed spin glasses, *Communications in Mathematical Physics* 405 no. 226 (2024)
33. D. Gamarnik, A. Jagannath, and A. Wein, Circuit lower bounds for the p -spin optimization problem, *Markov Processes and Related Fields (special issue for Inhomogeneous Random Systems 2022) 2024*, Vol. 30, No. 1, 81-96
32. D. Gamarnik, A. Jagannath, and A. Wein, Hardness of random optimization problems for boolean circuits, low-degree polynomials, and Langevin dynamics, *SIAM Journal on Computing* 53 no. 1 (2024) 1-46
31. G. Ben Arous, R. Gheissari, and A. Jagannath, High-dimensional limit theorems for SGD: effective dynamics and critical scaling, *Communications on Pure and Applied Mathematics* Vol. 77 (2024) pp. 2030-2080.
Conference version appeared at NeurIPS 2022 (Outstanding Paper Award, top 13 of $> 10,000$ submissions)
30. G. Ben Arous and A. Jagannath, Shattering versus metastability in spin glasses, *Communications on Pure and Applied Mathematics*, Vol 77, pp 139–176 (2024)

29. A. Baranwal, K. Fountoulakis, and A. Jagannath, Optimality of Message-Passing Architectures for Sparse Graphs, *Advances in Neural Information Processing Systems 37* (NeurIPS 2023), 2023
28. *K. Fountoulakis, A. Levi, S. Yang, A. Baranwal, and A. Jagannath, Graph attention retrospective, *J. Mach. Learn. Res.*, Vol 24, No. 246, 1-52 (2023) (Best paper award, ICLR GroundedML Workshop 2022) (authorship: contribution order)
27. A. Jagannath and N. Perkowski, A simple construction of the dynamical Φ_3^4 model, *Trans. Amer. Math. Soc.* Vol. 376, No. 3, pp 1507–1522 (2023)
26. A. Baranwal, K. Fountoulakis, and A. Jagannath, Effects of graph convolution on deep networks, *International Conference on Learning Representation 11* (ICLR 2023) (spotlight, top 5% of submissions), (2023)
25. D. Gamarnik, A. Jagannath, and S. Sen, The overlap gap property in principal submatrix recovery, *Probabability Theory and Related Fields* 181, pp 757–814 (2021)
24. A. Baranwal, K. Fountoulakis, and A. Jagannath, Graph convolution for semi-supervised classification: Improved linear separability and out-of-distribution generalization, *ICML 2021* (spotlight), *Proc. Mach. Learn. Res. Vol 139*, 684-693 (2021)
23. G. Ben Arous, R. Gheissari, and A. Jagannath, Online stochastic gradient descent on non-convex losses from high-dimensional inference, *Journal of Machine Learning Research*, Vol 22, No. 106, 1-51 (2021)
22. A. Jagannath and S. Sen, On the unbalanced cut problem and the generalized Sherrington-Kirkpatrick model, *Annales de l' Institut Henri Poincaré D, Combinatorics, Physics, and their Interactions* 8 (2021), pp 35-88
21. D. Gamarnik and A. Jagannath, The overlap gap property and approximate message passing algorithms for p -spin models, *Annals of Probability*, Vol 49, No. 1 (2021), 180-205.
20. D. Gamarnik, A. Jagannath, and A.S. Wein, Low-degree hardness of random optimization problems, *Proceedings of 2020 61st IEEE Symposium on Foundations of Computer Science (FOCS 2020)* pp. 131-140. Full version: arXiv:2004.12063v1
19. A. Jagannath, P. Lopatto, and L. Miolane, Statistical thresholds for tensor PCA, *Annals of Applied Probability* (2020), Vol. 30, No. 4, 1910-1933
18. G. Ben Arous, R. Gheissari, and A. Jagannath, Algorithmic thresholds for tensor PCA, *Annals of Probability*, Vol. 48, No. 4 (2020), 2052-2087
17. G. Ben Arous, R. Gheissari, and A. Jagannath, Bounding flows for spherical spin glass dynamics, *Communications in Mathematical Physics*, 373, 1011-1048 (2020)
16. A. Jagannath, Dynamics of mean field spin glasses on short and long timescales, *Journal of Mathematical Physics* 60, 083305 (2019) (special issue for ICMP 2018)
15. A. Auffinger and A. Jagannath, Thouless–Anderson–Palmer equations for generic p -spin glasses, *Annals of Probability* Vol. 47 No. 4 (2019), pp 2230–2256
14. R. Gheissari and A. Jagannath, On the spectral gap of spherical spin glass dynamics, *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques*, Vol. 55, No. 2 (2019), 756-776.
13. A. Auffinger and A. Jagannath, On spin distributions for generic p -spin models, *Journal of Statistical Physics* (2019) 174: 316
12. G. Ben Arous and A. Jagannath, Spectral gap estimates for mean field spin glasses, *Communications in Mathematical Physics* (2018), Vol. 361, No. 1, pp 1-52

11. A. Jagannath, J. Ko, and S. Sen, MAX κ -CUT and the inhomogeneous Potts spin glass, *Annals of Applied Probability*, Vol. 28, No. 3 (2018), pp 1536-1572
10. A. Jagannath and I. Tobasco, Bounding the complexity of replica symmetry breaking for spherical spin glasses, *Proceedings of the American Mathematical Society*, Vol. 146, No. 7, (2018) pp 3127-3142
9. A. Jagannath and T. Trogdon, Random matrices and the New York City subway system, *Physical Review E* 96, 030101(R), (2017)
8. A. Jagannath and I. Tobasco, Low temperature asymptotics in spherical mean field spin glasses, *Communications in Mathematical Physics*, Vol. 352, No. 3 (2017), pp 979-1017
7. A. Jagannath and I. Tobasco, Some properties of the phase diagram of mixed p -spin glasses, *Probability Theory and Related Fields* Vol 167 (2017) pp 615-672.
6. A. Jagannath, Approximate ultrametricity for random measures with applications to spin glasses, *Communications on Pure and Applied Mathematics* Vol 70 (2017), 611-664.
5. A. Jagannath, On the overlap distribution of branching random walks, *Electronic Journal of Probability*, Vol. 21. Paper No. 50, (2016) 16 pp.
4. A. Jagannath and I. Tobasco, A dynamic programming approach to the Parisi functional, *Proceedings of the American Mathematical Society*, Vol. 144, No. 7, (2016) pp 3135-3150
3. A. Jagannath, Variational and structural methods in mean field spin glasses, Ph. D. Thesis, May 2016
2. S. Heilman, A. Jagannath, and A. Naor, Solution of the propeller conjecture in $\mathbb{R}^3$, *Discrete and Computational Geometry*, Vol. 50, No. 2 (2013), pp 263-305. (an extended abstract appeared at STOC 2012.)
1. A. Jagannath and H. Weitzner, Charged particle motion in electromagnetic fields varying moderately slowly in space, *Physics of Plasmas* 18, 104510 (2011).

Invited talks and minicourses

Invited Minicourses at Summer Schools

3. High-dimensional probability and its applications, Northwestern University, Aug 2026
2. Statistical Mechanics of Spin Glasses, Neural Networks, and Learning, Hausdorff School for Mathematics, Universitat Bonn, Bonn, June 2025
1. Online Open Probability School (prev. Séminaire de Mathématiques Supérieures, Centre de Recherche Mathématiques, June 2020

Invited Talks

52. Center for Data Science Colloquium, NYU, Apr 2026
51. Statistical Physics and Machine Learning Back Together Once Again, Institut d'Études Scientifique de Cargèse, Aug 2025
50. Phase transitions and dynamics in random media, McGill/CRM, Montreal, June 2025
49. Tutte Colloquium, University of Waterloo, Waterloo, April 2025
48. PIMS-UBC Distinguished Mathematics Colloquium, PIMS and UBC, Vancouver, Feb 2025
47. Probability Seminar, University of British Columbia, Vancouver, Feb 2025

46. Recent developments beyond Classical Regimes in Statistical Learning, Toulouse, Nov 2024
45. 2nd ICML Workshop on High-dimensional Learning Dynamics, ICML, Vienna, July 2024
44. Computational complexity of Statistical Inference, BIRS, Feb 2024
43. Mathematics of Data Science, National University of Singapore, Singapore, Jan 2024
42. Machine Learning Theory Workshop, Vector Institute and University of Waterloo, Nov 2023
41. Analytical Approaches for Neural Network Dynamics. Institut Henri Poincaré, Paris, Oct 2023
40. High Dimensional Statistics and Random Matrices, Porquerolles, June 2023
39. CanaDAM, Winnipeg, June 2023
38. Optimization in data science, SIAMOPT, Seattle May 2023
37. Mossel-Sun working group, MIT, April 2023
36. Probability seminar, MIT, April 2023
35. Topics in High Dimensional Probability, ICTS-TIFR, Bangalore, Jan 2023
34. Spin glass workshop, SwissMAP research centre, Les Diableret, Sept 2022
33. Inhomogeneous Random Systems 2022 (29th annual), Institut Curie and Institut Henri Poincaré, Paris, Jan 2022
32. Probability Seminar, University of Toronto, Dec 2021
31. Session on "Mathematical Foundations of Machine Learning", Canadian Mathematical Society Winter Meeting, Vancouver, Dec 2021
30. Montreal probability seminar, Centre de Recherche Mathématiques and McGill, Oct 2021
29. Statistics seminar, U.C. Davis, April 2021
28. Algorithms and Complexity seminar, University of Waterloo, September 2020
27. Youth in High Dimensions, International Centre for Theoretical Physics, July 2020
26. Mathematical Methods and Models in Machine Learning, University of Bologna, April 2020
25. The rough high-dimensional landscape problem, Kavli Institute for Theoretical Physics, Feb. 2019
24. Probability seminar, MIT, November 2018
23. Math, Information, and Computation seminar, NYU Center for Data Science, November 2018
22. Math and Data working group, NYU Center for Data Science, November 2018
21. Stochastics and Statistics seminar, Institute for Data, Systems, and Society, MIT, October 2018
20. Spin glasses and related topics, Banff International Research Station, October 2018
19. Probability seminar, Cornell University, September 2018
18. Scaling limits in models of statistical mechanics, Oberwolfach, September 2018
17. Statistical physics and machine learning back together, Institut d'Etudes Scientifique de Cargese, August 2018
16. Session on Equilibrium statistical mechanics, International Congress of Mathematical Physics, Montreal, July 2018
15. Simons meeting on High-dimensional dynamics, Simons Collaboration on "Cracking the Glass Problem", CUNY, April 2018 (Keynote speaker)

14. Dynamical systems seminar, Lefschetz Center, Brown University, April 2018
13. Probability seminar, University of Toronto, January 2018
12. First joint meeting of CRM-IMPA, Centre de Recherche Mathematiques, July 2017
11. Phase transitions in random computational problems, American Institute of Mathematics, June 2017
10. Analysis seminar, University of Toronto, March 2017
9. Probability seminar, Stanford University, February 2017
8. Analysis and probability seminar, University of Michigan, February 2017
7. Random matrix theory and probability seminar, Harvard University, February 2016
6. Probability seminar, Northwestern University, February 2016
5. Probability and statistical physics Seminar, University of Chicago, January 2016
4. Probability seminar, Cornell University, September 2015
3. Analysis Seminar, Northwestern University, June 2015
2. Doctoral student working group seminar of LPMA (Paris VI and VII), April 2015
1. Spin glasses and related topics, Banff International Research Station, July 2014

Teaching, Advising, and Professional Activities

Advising

PhD Students

Xiwei Tian (PhD in Statistics, U. Waterloo, 2025 –)
 Nigel Petersen (PhD in Statistics, U. Waterloo, 2025–)
 Varnan Sarangian (PhD in Statistics, U. Waterloo, 2024–)
 Taj Jones-McCormick (PhD in Statistics, U. Waterloo, 2023–)
 Aseem Baranwal (PhD in CS U. Waterloo, 2024 → XTY Labs)

Postdocs

Curtis Grant (2026–)
 Han Le (2025–)
 Justin Ko (2023-2025 → Asst. Prof. Syracuse U.)
 Yiming Xu (2023-2024 → Asst. Prof. U. Kentucky)

Master's Students

Lily Seebach (MMath in Data Science, U. Waterloo, 2026 → Data scientist at BMO)
 Parsa Rangriz (MMath in Statistics, U. Waterloo, 2025 → PhD at UCSD)
 Victor Issa (Masters' Intern, ENS Lyons, 2023 → PhD ENS Lyons)
 Tingzhou Yu (MMath in Statistics, U. Waterloo, 2023 → PhD at U. Alberta)
 Philipp Schröppel (MMath in Statistics, U. Waterloo, 2022 → Industrial PhD at Ulm U. CS)
 Anna Brandenberger (MSc in Physics, Perimeter Institute 2022 → PhD at MIT Math)

Teaching

University of Waterloo

Instructor: Stat 230 – Probability Theory, Fall 2021; Stat 240 – Probability Theory (advanced level), Fall 2019; Stat 333 – Stochastic Processes 1, Winter 2020, Winter 2022; Stat 901 – Theory of Probability 1 (Graduate), Fall 2021; Stat 902 – Theory of Probability 2 (Graduate), Winter 2024; Stat 903 – High dimensional probability, Winter 2026 Stat 946 – Math of Data Science (advanced topics), Winter 2021, Stat 946 – Mathematical Foundations of Deep Learning, Fall 2024;

PhD proposal committee: Takaai Koike (2019), Weinan Qi (2020), Kelly Ramsay (2020), Hongda Hu (2022)

PhD Defense: Sina Baghal (2021), Kelly Ramsay (2022), Jacob Campbell (2023), Tosca Lechner (2024)

External Examiner: Fu-Husan Ho (PhD in Math, Toulouse, 2023), Tomas Dominguez (PhD in Math, University of Toronto, 2024), Vanessa Piccolo (PhD in Math, ENS Lyons, 2025), Victor Issa (PhD, ENS Lyons, 2026)

MMath Thesis committee: Varnan Sarangian (2023)

Harvard University

Instructor: Math 21a – Multivariable calculus (Undergraduate), Fall 2017; Topics In Mean Field Spin Glasses (Graduate), Spring 2018, Probability Theory, Spring 2019

PhD Defense: Benjamin Landon (chair and reader, 2018), Zilian Che (chair and reader, 2018)

New York University

Recitation Leader: Theory of Probability (Undergraduate), Spring 2014; Honors Analysis 1 (Undergraduate), Fall 2014, Fall 2015

Grader: Adv. Topics in Probab.: Random Graphs (Graduate), Spring 2016

Editorial work

Mathematical Statistics and Learning, Associate Editor, 2024–

Seminar and Conference Organization

Principal Organizer, High-Dimensional Learning Dynamics (26w5565), BIRS (Banff), Winter 2026

Organizer, 3rd Workshop on High-dimensional learning dynamics, ICML 2023, Vancouver, 2025

Founder and chair, Waterloo Probability Seminar, Winter 2020–

Organizer, Analysis of Complex Data: Tensors, Networks, and Dynamic Systems (24w5263), BIRS (Banff), Spring 2024

Organizer, Harvard-CMSA Random Matrix and Probability Theory Seminar, Fall 2017– Spring 2019

Organizer, Brandeis-Harvard-MIT-Northeastern Colloquium, Fall 2017– Spring 2018

Organizer, Toronto Probability Seminar, Fall 2016– Spring 2017

Organizer, Special Session on Spin Glasses and Disordered Media, JMM (AMS), Jan. 2017

Organizer, Courant Graduate Student Post-Doc Seminar (NYU) Spring 2014–Fall 2015

Organizer, Student Probability Seminar (NYU) Fall 2013–Spring 2016

Last updated: August 28, 2026